

Venkatesh

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Senior Data Platform / Cloudera Engineer with **12+ years of experience** designing, operating, and hardening **enterprise-scale Cloudera Data Platform (CDP – Private Cloud Base)** and **Cloudera Machine Learning (CML) on Kubernetes**. Specialized in building **resilient, high-availability data and ML platforms** through proactive memory tuning, resource governance, and workload optimization. Strong background in root cause analysis, platform standardization, and mentoring teams to achieve long-term operational independence in large, multi-tenant environments.

CORE SKILLS

Cloudera Data Platform (CDP) Administration, Big Data Administration, Cloud Deployment, Cluster Management, Performance Optimization, Data Security and Governance, Data Migration, High Availability and Disaster Recovery, Backup and Recovery, DevOps Engineering

Professional Experience

NucleusTeq, Bengaluru, IN
Technical Lead

May 2024 to Present

- Own and operate **enterprise Cloudera CDP Private Cloud environments** supporting large-scale data engineering and machine learning workloads.
- Stabilized data and ML platforms by identifying systemic performance bottlenecks across **HDFS, YARN, Spark, and CML**, improving overall reliability and availability.
- Optimized **CML on Kubernetes** by tuning CPU/memory requests, limits, and namespace quotas, significantly reducing job failures and improving cluster utilization.
- Led proactive **memory and resource optimization** for Spark and ML workloads, aligning JVM configuration with container-level constraints to prevent runtime failures.
- Optimized large-scale Spark batch pipelines by tuning executor sizing, memory overhead, shuffle behavior, and partition strategies, improving throughput and reducing job failures in production Cloudera environments.
- Designed and validated **High Availability (HA)** configurations and removed single points of failure across critical platform components.
- Performed deep **root cause analysis (RCA)** on production incidents using logs, metrics, and workload patterns; implemented durable fixes to prevent recurrence.
- Improved **HDFS capacity planning and data lifecycle management**, reducing storage pressure and operational risk.
- Defined **platform standards, best practices, and operational runbooks**, enabling internal teams to independently manage and troubleshoot the platform.
- Acted as a **technical mentor and escalation point** for platform, data engineering, and ML teams.

Alation, Bengaluru, IN
Staff Technical Engineer

May 2023 to February 2024

- Served as platform owner for enterprise Cloudera Data Platform (CDP) **deployments across AWS, Azure, and on-premises, ensuring high availability, scalability, security, and seamless hybrid-cloud integration.**
- Acted as escalation point for complex production issues, **conducting in-depth RCA for critical outages, driving permanent fixes, and embedding fault-tolerant designs to improve platform resilience.**
- Led capacity planning, infrastructure scaling, and version upgrades, **automating operational workflows with Shell scripting and Ansible to standardize deployments, reduce operational risk, and improve efficiency.**

- Designed and operationalized observability and SRE practices, **implementing Grafana and Prometheus-based monitoring, alerting, and performance baselining to enable proactive issue detection and rapid recovery.**
- Championed platform governance, security, and operational maturity, **developing SOPs and runbooks for Hadoop/CDP operations, enforcing DevOps and ITIL alignment, and implementing Kerberos- and Ranger-based security controls with cross-functional teams.**

Acceldata Technology Pvt Ltd, Bengaluru, IN

April 2021 to May 2023

Solutions Engineer

- Spearheaded enterprise Hadoop modernization initiatives as part of **Acceldata's Open Data Platform (ODP)**, executing zero-downtime migrations from legacy distributions to ODP Hadoop, resulting in ~30% improvement in processing efficiency and platform stability
- **Established technical stewardship for distributed data platforms** (Hadoop, Kafka, HBase) across cloud and hybrid environments, shaping architecture, scalability, and resilience strategies.
- **Championed production reliability and incident prevention**, leading complex root cause analyses, driving systemic remediation, and reducing repeat incidents by approximately 30%.
- **Institutionalized operational excellence practices**, automating backup and recovery using Shell and Python, implementing Pulse-based observability and alerting, advancing cluster hardening beyond L3 benchmarks, and standardizing SOPs and DevOps runbooks across teams.

PRIOR WORK EXPERIENCE

Technology Services Specialist, IBM, Bengaluru, IN, Apr 2017 to February 2021

IT Analyst, TCS, Bengaluru, IN, September 2015 to February 2017

IT Analyst, Future Focus Pvt Ltd (Deployed to TCS), Bengaluru, IN, February 2015 to September 2015

Senior Associate, Genpact, Bengaluru, IN, June 2011 to February 2015

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science (B.Sc.) from Sri Venkateswara University, 2008

AWS Certified Solutions Architect - Associate

AWS Certified Machine Learning - Specialty

TECHNICAL SKILLS

- **Big Data & Hadoop Technologies**
 - Hadoop Ecosystem: HDFS, YARN, ZooKeeper, Hive, Spark, Impala, HBase, Kudu
 - Security & Governance: Kerberos, Ranger
 - Cloud Integration: AWS
 - Hadoop Distributions: Cloudera CDH, Cloudera CDP, Hortonworks HDP, ODP
 - Performance Tuning and Optimization
 - Cluster Management: Installation, Configuration, Monitoring
- **Programming & Scripting Languages:** Apache Spark (Scala), SQL, Python, Unix Shell Scripting
- **Cloud Platforms: Amazon Web Services (AWS):** EMR, S3, EC2, Microsoft Azure: HDInsight, Google Cloud Platform (GCP): Dataproc, BigQuery
- **Database Technologies:** PostgreSQL (EDB), Postgres Enterprise Manager, Oracle Database, Backup and Recovery
- **Operating Systems:** Linux, Windows